

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- **Recitation**: Tuesdays @6 pm in Keck Annex
- Due **this Friday, 8/30 @5 pm**:
  - If you find a group, please self-enroll on Brightspace -> Groups (all members in your group need to do so)
  - After the deadline, students who have not signed up for a group will be randomly assigned to a group.
- Due next Friday, 9/6 @5 pm:
  - Classwork 1

Last Lecture

- Human Intelligence vs. Artificial Intelligence
- AGI vs. ASI
- Levels of Intelligence
  - Associational (L1)
  - Interventional (L2)
  - Counterfactual (L3)

This Lecture

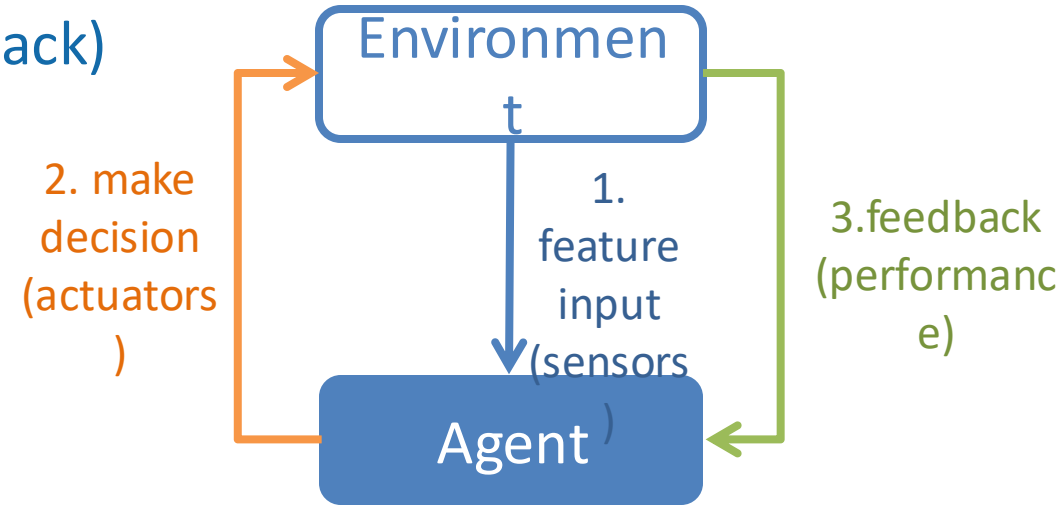
- Artificial and Logical Agents
- Propositional Logic
  - Propositional Syntax
  - Propositional Semantics

Artificial Agents

- What is an agent?
- An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and acting upon that environment through **actuators**.
- Examples?
  - Human agent
  - Robotic agent
  - Software agent
  - …

How an agent operate?

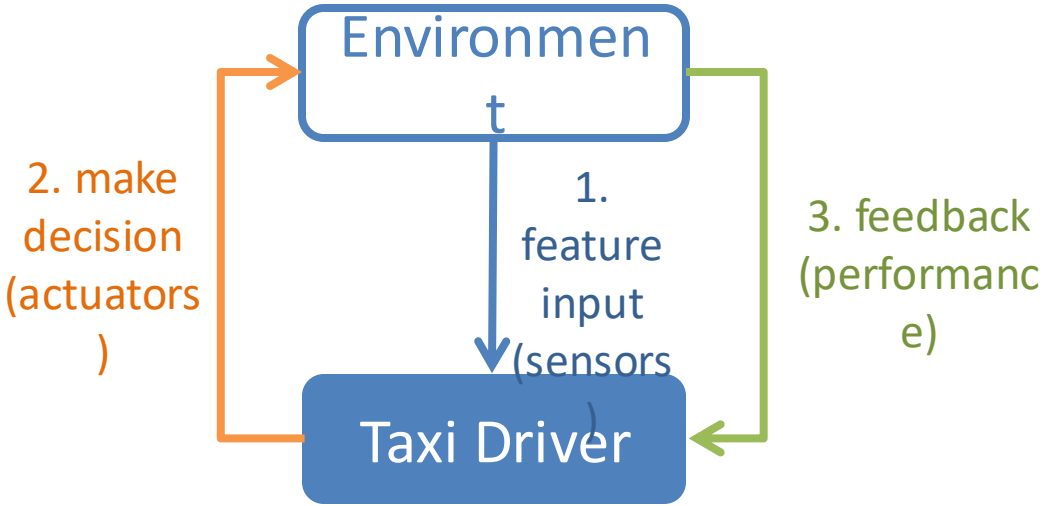
- PEAS Description
  - Performance (Feedback)
  - Environment
  - Actuators
  - Sensors



More Examples

- **Taxi Driver**
  - Performance
  - Environment
  - Actuators
  - Sensors

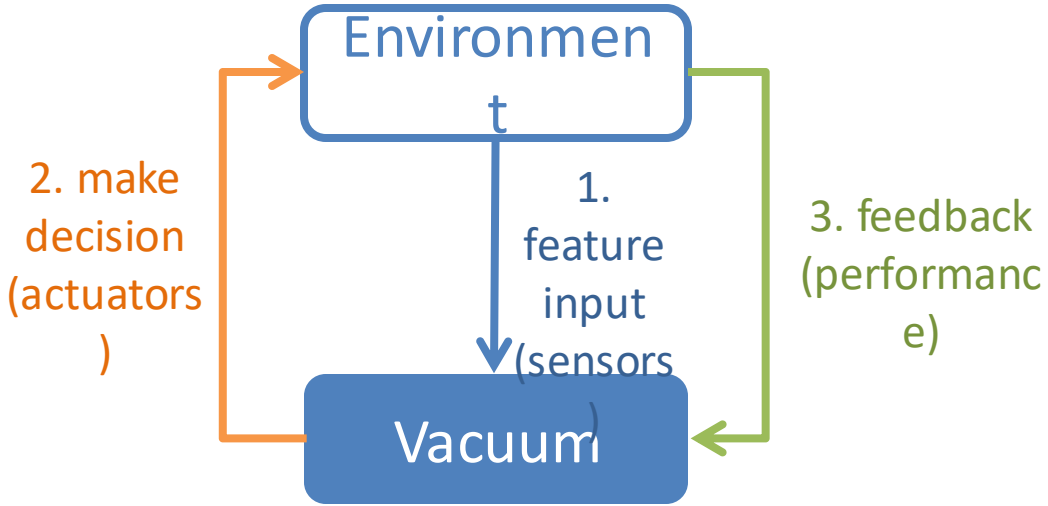




More Examples

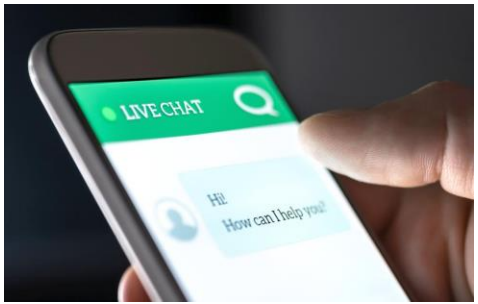
- **Robotic Vacuum Cleaner**
  - Performance
  - Environment
  - Actuators
  - Sensors

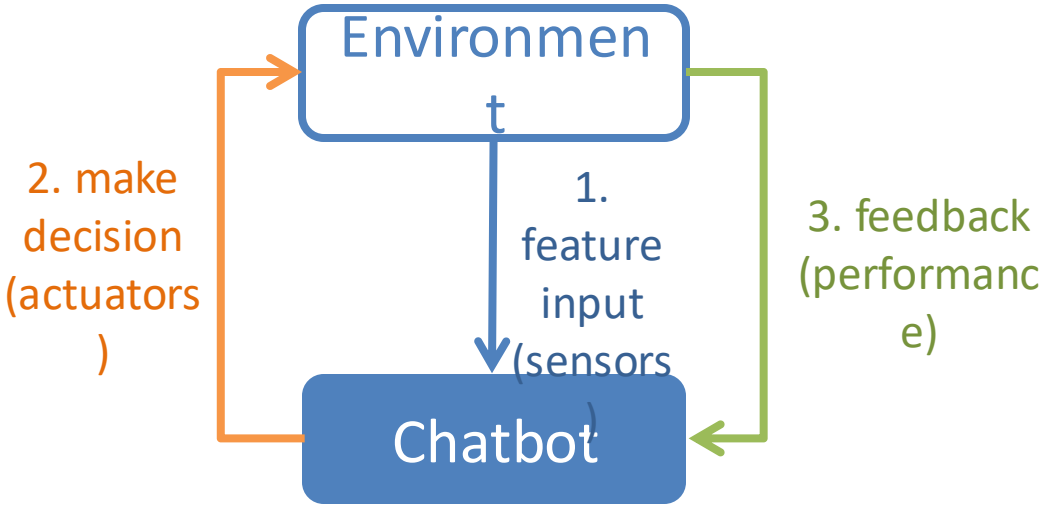




More Examples

- **Customer Service Chatbot**
  - Performance
  - Environment
  - Actuators
  - Sensors





Logical Agents

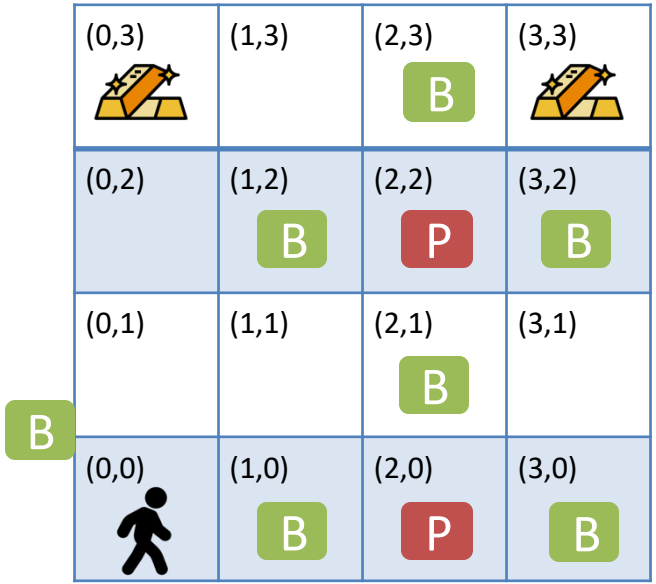
- A logical agent is a type of artificial agent that uses **logical reasoning** to **make decisions** and **take actions**.
- What is logical reasoning?

Inference

- Inference is the ability to take **knowledge** about the mechanics of an environment (**rules**), marry those to **observations** about the environment (**facts**), and then **deduce** new facts as a result (**inferences**).
- Example:
  - Rule: All humans are mortal.
  - Facts: Socrates is a human.
  - Inference: Socrates is mortal.

The (Simplified) Wumpus World Example

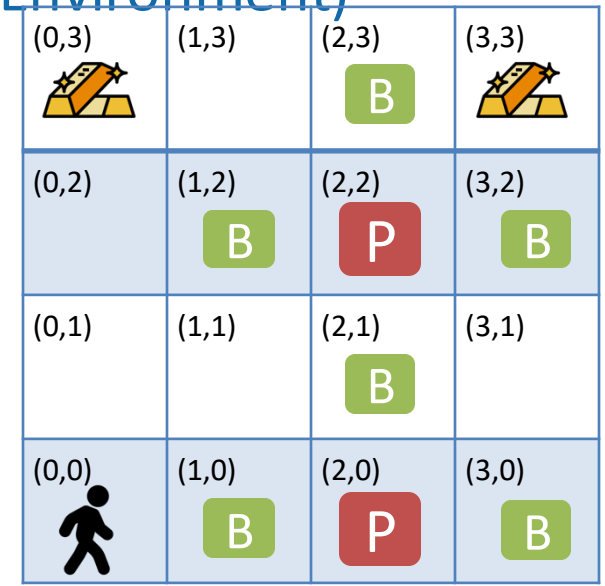
- Grid world, starting from (0,0)
- Find all gold in the grid (safely)
- Features dangerous pits P
- Each pit is surrounded by breezes



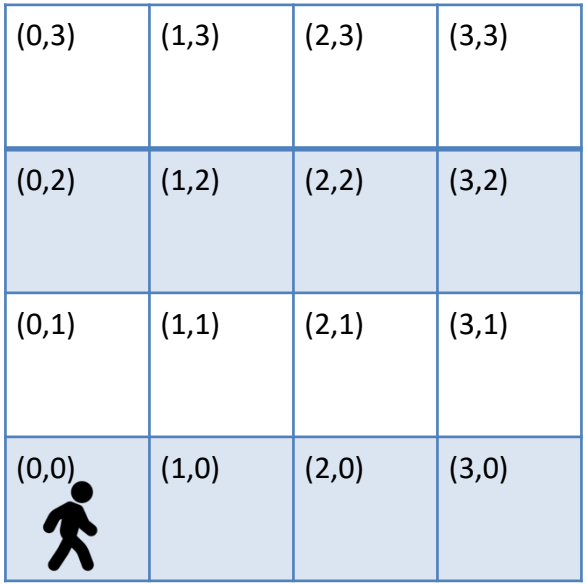
Objective: find all gold w/o stepping into any pit.

Environment vs. Agent

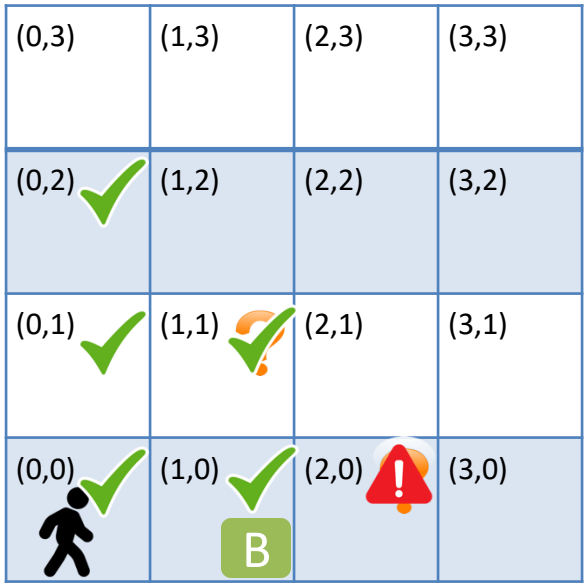
- Real/True World (Environment)



- **Agent's** Representation



How to solve it?



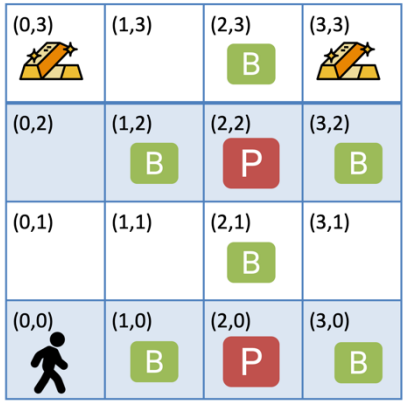
P Pit

B Breeze

Known Safe

Possible Pit

Deduced/Known Pit



Let’s try it out

- https://luka1199.github.io/wumpus-world/

How to formalize the logic?

- Goal: develop **inference engine** that can automatically **combine rules/facts** and **make deductions** about their environments

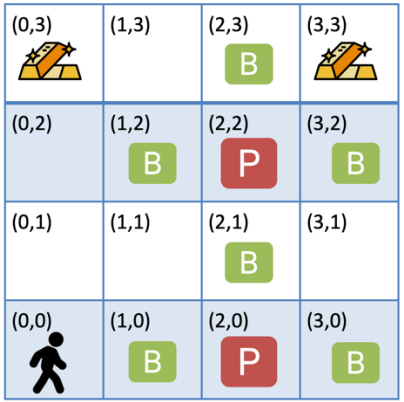
Knowledge Base

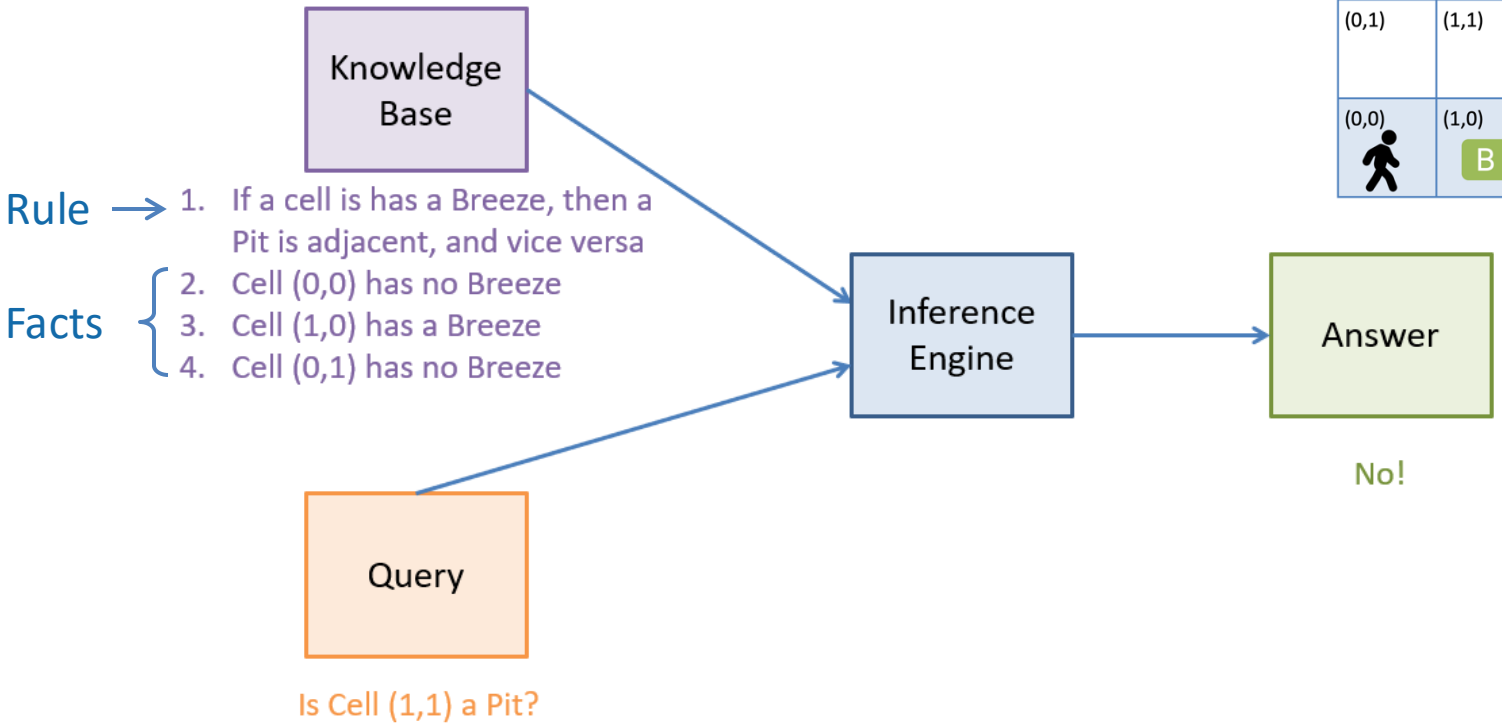
- **Knowledge Base (KB)** consists of logical sentences that represent assertions about the world.
  - **Rules**: describe how the world works including any **background knowledge** (the initial state of the KB)
    - E.g., All humans are mortal
  - **Facts**: describe **observations** made by the agent about the environment (updated online as it interacts).
    - E.g., Socrates is a human.

Inference Engine

- The process of inference is handled by an **inference engine**, which is parameterized by
  - KB: a knowledge base
    - Rule: All humans are mortal.
    - Fact: Socrates is a human.
  - Inference procedure: produces an answer to a given query
    - Inference: Socrates is mortal.

Inference Process





Propositional Logic

- Propositional logic is a knowledge representation language (e.g., KB) in which **propositions** (T/F variables) make **assertions** about aspects of the reasoning environment.
- Assertions
  - Rules: general truths
  - Facts: experientially derived

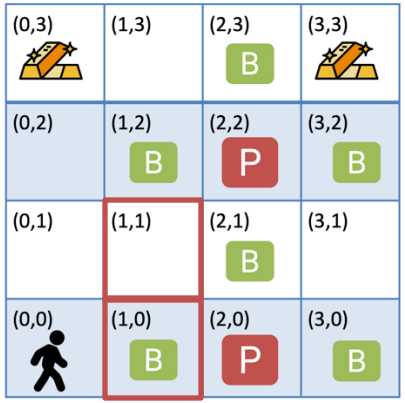
Propositional Syntax

- The **syntax** of propositional logic defines what constitutes a logical expression.
- **Propositional symbols** are "placeholders" (Boolean variables) for truth values, and include two "constant” symbols:
  - **True** or **T** to denote the "always true" proposition.
  - **False** or **F** to denote the "always false."

Propositional Syntax

- Example
  - $B_{1,0}$: There is a breeze in (1,0)
  - $\neg P_{1,1}$: There is NO pit in (1,1)

Logical Negation



- **Atomic** sentences are propositional sentences containing only a **single** propositional symbol.

Propositional Syntax

- **Complex** sentences are constructed from simpler sentences using logical grouping **()** and **connectives**.

Symbol	Name	Interpretation
¬ (not)	Negation	$\neg\alpha$ indicates that the truth value of sentence α is flipped. An atomic sentence or negated atom is called a **literal.**
∧ (and)	Conjunction	$\alpha \wedge \beta$ indicates the logical "and" operator between two sentences α, β. A sentence composed of only conjoined literals is called a **term**.
∨ (or)	Disjunction	$\alpha \vee \beta$ indicates the logical "or" operator between two sentences α, β. A sentence composed of only disjoined literals is called a **clause**.
⇒ (implies)	Implication	$\alpha \Rightarrow \beta$ indicates that sentence α (the **premise**) "implies" sentence $beta$ (the **conclusion**). Implications are propositional syntax for **if-then** statements.
⇔ (if and only if)	Biconditional	$\alpha \Leftrightarrow \beta$ indicates that α and β mutually imply one another.

Quick Summary

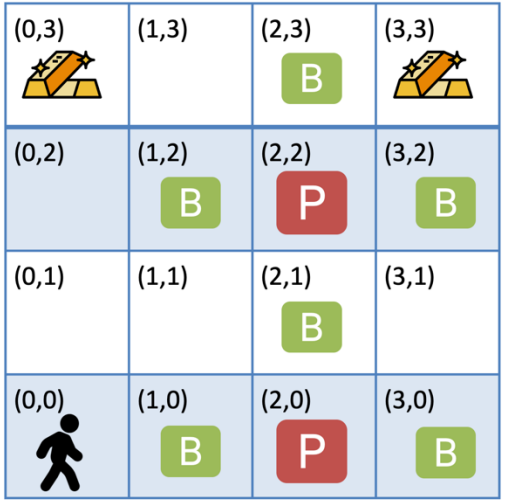
$$
\begin{aligned}
\mathit{Sentence} &\rightarrow \mathit{AtomicSentence} \mid \mathit{ComplexSentence} \\
\mathit{AtomicSentence} &\rightarrow \mathit{True} \mid \mathit{False} \mid P \mid Q \mid R \mid \ldots \\
\mathit{ComplexSentence} &\rightarrow (\ \mathit{Sentence}\ ) \mid [\ \mathit{Sentence}\ ] \\
&\mid \neg\ \mathit{Sentence} \\
&\mid \mathit{Sentence} \wedge \mathit{Sentence} \\
&\mid \mathit{Sentence} \vee \mathit{Sentence} \\
&\mid \mathit{Sentence} \Rightarrow \mathit{Sentence} \\
&\mid \mathit{Sentence} \Leftrightarrow \mathit{Sentence}
\end{aligned}
$$

$$
\textsc{Operator Precedence} \quad : \quad \neg, \wedge, \vee, \Rightarrow, \Leftrightarrow
$$

Practice

Translate each of the following sentences to logical ones. Also denote whether it is a rule or a fact.

- There is no Breeze at (0,0)
- There is no Breeze at (0,0) nor is there a Pit at (0,1)
- There is a Breeze in (1,0), if and only if there is a pit in an adjacent (legal) cell



¬ (not)	Negation
∧ (and)	Conjunction
∨ (or)	Disjunction
⇒ (implies)	Implication
⇔ (if and only if)	Biconditional

Propositional Semantics

- What logical sentence "mean"
- A **world** w is a possible instantiation of all propositions (i.e., an assignment of **`True`**, **`False`** to all props.)
  - Suppose we have 2 propositions, P,Q, how many possible worlds are there?
- **Propositional semantics** define the rules that decide whether a given sentence is **`True`** or **`False`** in some world.

Truth Table

World	P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
w_1	F	F	T	F	F	T	T
w_2	F	T	T	F	T	T	F
w_3	T	F	F	F	T	F	F
w_4	T	T	F	T	T	T	T

Remaining of the lecture

CLASSWORK 1